MIL-E-48223A (PA)
30 December 1974
SUPERSEDING
MIL-E-48223 (MU)
31 August 1973

MILITARY SPECIFICATION

ELECTROCOULOMETER, DUAL ANODE, TWO MICROAMPERE HOURS AND NINE HUNDRED AND SIXTY MICROAMPERE HOURS CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the quality and product assurance acceptance and qualification requirements for one type of dual anode electrocoulometer (two microampere hours capacity on one anode and nine hundred and sixty microampere hour capacity on the other anode).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Prcedures and Tables for Inspection by Attributes

FSC: 5999

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MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests For
MIL-STD-1167	- Ammunition Data Card
MIL-STD-1168	- Lot Numbering of Ammunition
MIL-STD-1169	- Packaging, Packing and Marking For Shipment for Inert Ammunition Components.
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

9276886 - Dual Anode E-Cell Coulometer

(Copies of specifications, standards, drawings and publications listed above required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable specifications and drawings.

3.2 Part. - The part shall comply with all requirements specified on drawing (dwg.) 9276886 and with all requirements specified in applicable specifications.

3.3 Electrical Requirements

3.3.1 Electrical Capacity

3.3.1.1 Anode No. 1. - Anode No. 1 to the cathode shall not develop a voltage greater than one hundred and seventy-five (175) millivolts across it during the first sixty (60) seconds of operation while a current of eighty plus or minus four tenths of one percent (80 $\pm$ 0.4%) micro amperes direct current flows through it at all temperatures of zero degrees centigrade (0°C)

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to seventy one degrees centigrade (71°C). Anode No. 1 to the cathode shall not develop a voltage drop greater than two hundred (200) millivolts across it during the first sixty (60) seconds of operation while a current of eighty plus or minus four tenths of one percent (80 ± 0.4%) microamperes direct current flows through at all temperatures from zero degrees centigrade to minus thirty two degrees centigrade (0°C to -32°C). The voltage drop (at any temperature) shall increase to a minimum of eight hundred (800) millivolts within a maximum time of one hundred and twelve (112) seconds without changing the eighty (80) microampere current.

3.3.1.2 Anode No. 2. - Anode No. 2 to the cathode shall not develop a voltage drop greater than one hundred and seventy-five (175) millivolts across it for the first thirty-seven (37) hours of operation while a current of twenty-one plus or minus four tenths of one percent (21 ± 0.4%) microamperes direct current flows through it at all temperatures between zero degrees centigrade to seventy-one degrees centigrade (0°C to 71°C). Anode No. 2 to the cathode shall not develop a voltage drop greater than two hundred (200) millivolts across it during the first thirty-seven (37) hours of operation while a current of twenty-one plus or minus four tenths of one percent (21 ± 0.4%) microamperes direct current flows through it at all temperatures between minus thirty-two degrees centigrade to zero degrees centigrade (-32°C to 0°C). The voltage drop (at any temperature) shall increase to a minimum of eight hundred (800) millivolts to some time between forty one (41) and one tenths to forty five and seven tenths (4.1.1 to 45.7) hours without changing the twenty-one (21) microampere current.

3.4 Mechanical and Environmental Test

3.4.1 Terminal Strength

3.4.1.1 Cathode Lead Weld

3.4.1.1.1 Pull Test. - The cathode lead to case weld shall not separate when pulled with a minimum axial load of twenty-four (24) pounds.

3.4.1.1.2 Bend Test. - The cathode lead to case weld shall not separate when the lead is bent a minimum of three times (3x) with a two (2) pound minimum load in a ninety degree (90°) angle from the lead axis.

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3.4.1.2 Anode No. 2 lead weld

3.4.1.2.1 Pull Test. - The anode lead to anode weld shall not separate when pulled with a minimum axial load of twenty-eight (28) pounds.

3.4.1.2.2 Bend test. - The anode lead to anode weld shall not separate when the lead is bent a minimum of three times (3x) with a two (2) pound minimum load in a ninety degree (90°) angle from the lead axis.

3.4.2 Hermetic Seal. - The electrocoulometer shall meet a hermetic seal leakage requirement of 1 x 10 -6 CC/second of helium after being subjected to terminal strength and thermal shock tests. The terminal strength shall be tested in accordance with the requirements of 3.4.1 with the capacity test excluded. The electrocoulometer shall be thermal shocked by cycling five (5) times to the following schedule.

a. Store at minus fifty-five degrees centigrade (-55°C) maximum for thirty plus five (30 + 5) minutes.

b. Transfer within five (5) minutes to plus seventy-one degrees centigrade (+71°C) minimum for thirty plus five (30 + 5) minutes.

c. Transfer within five (5) minutes back to minus fifty five degrees centigrade (-55°C).

The five cycles shall consist of repeating a through b five times (5x).

3.4.3 Extreme Temperature Storage. - The electrocoulometer shall be tested in accordance to the requirements of Test Method 112 of MIL-STD-331 for extreme temperature storage. The device shall be tested for the capacity requirements of 3.3.1 at a temperature of twenty-five degrees centigrade (25°C) after the completion of the storage cycle.

3.4.4 Test Encapsulation. - The electrocoulometer shall be encapsulated where required, in a potted configuration of one (1) inch length with a cross section of one half (1/2) inch square. The leads shall be exposed and not damaged. The encapsulation material shall be made from Product Research Company potting compound No. PR-1547 or equivalent.

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3.4.5 Mechanical Shock. - The electrocoulometer shall be encapsulated as required by 3.4.4 and mechanically shocked in accordance with the following schedule:

a. Magnitude: Thirteen thousand (13,000) G's peak minimum.
b. Pulse Duration: Three tenths (0.3) millisecond plus or minus twenty percent (± 20%) at ten percent (10%) of peak amplitude points.
c. Sine Shape: Approximately one half (1/2) sine wave.
d. Direction: Shock shall be in one (1) plane only for a test position. The device shall be tested for the capacity requirements of 3.3.1 at a temperature of plus twenty-five degrees centigrade (+ 25°C) after the completion of the shock tests.

3.4.6 Vibration. - The electrocoulometer shall be encapsulated as required by 3.4.4 and vibrated in accordance with Test Method 204A of MIL-STD-202 with the following exceptions:

a. Paragraph 4.5 shall not apply.
b. Electrocoulometer shall be non-operating during vibration.
c. Vibration per one mounting position shall be in one direction only.

The electrocoulometer shall meet the capacity requirements of 3.3.1 at a temperature of plus twenty-five degrees centigrade (+25°C) after the completion of the vibration tests.

3.4.7 Altitude. - The electrocoulometer shall be tested for the altitude requirements of Test Condition B, Test Method 105C of MIL-STD-202 for one (1) hour minimum. The device shall be tested for the requirements of 3.3.1 at a temperature of plus twenty-five degrees centigrade (+25°C) after completion of the altitude tests.

3.5 Marking and Identification. - All test units shall be coded to identify them at all times throughout the test program.

3.6 Test Position. - Unless specified otherwise, test units in a specific group shall be evenly divided into equal groups and tested as follows:

One third (1/3) in a horizontal position.

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One third (1/3) in a vertical position with the anode leads directed upward.

One third (1/3) in a vertical position with the anode leads directed downward.

3.7 Seals

3.7.1 Cathode. - The cathode lead and cup shall not show evidence of electrolyte leakage, etching, discoloration due to electrolyte leakage or evidence of foreign substance(s).

3.7.2 Anode. - The seal and seal area shall not show evidence of electrolyte leakage, discoloration due to electrolyte leakage, evidence of foreign substance(s) or seals not seated in the cup crimp.

3.8 Test Condition. - All examinations and tests shall be made under the following conditions unless specified otherwise:

Temperature: 25°C ± 3°C
Humidity: 20% Relative Humidity (Hg) + 40% RH
Pressure: 29 Inches of Mercury (Hg) ± 2 inches Hg.

3.9 Workmanship. - All electrocoulometers shall be assembled and fabricated in a thorough, workmanlike manner. They shall be uniformily free of sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign material. Cleaning methods used shall not be injurious to the device, nor shall the device be contaminated by any cleaning agent. All markings and stampings shall be neat and clearly defined.

3.10 Submission of Results of Contractor Conducted Examinations. - The contractor shall submit the results of his examinations or tests to the designated address as required by the contract. Results shall be submitted within thirty (30) days from the completion of each lot tested. (See 6.4).

3.11 First Article Testing. - This specification makes provision for first article testing. Submission of the first article sample quantity by the contractor shall be as specified in the contract. (See 4.2)

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the suppler is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product. - At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot
e. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subseqent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2 (See 6.1).

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The first article sample shall consist of five (5) sets of cathode cup - lead assemblies and total of five hundred and thirty one (531) electrocoulometers for his first article samples. All items submitted in the first article sample shall have been produced by the contractor using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and shall submit a record of this inspection with the sample, including statements of findings of tests and examinations and certificates of conformance for materials (See 4.1.1.c). A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. - First article testing and inspections shall be performed to the following requirements.

a. Fifty (50) electrocoulometers shall be inspected for the requirements of 3.7.1, 3.7.2 and 4.3.2.1.
b. Three (3) cathode cup - lead assemblies shall be tested for the requirements of 3.4.1.1.1.
c. Two (2) cathode cup lead assemblies shall be tested for the requirements of 3.4.1.1.2
d. Three (3) electrocoulometers shall be tested for the requirements of 3.4.1.2.1.
e. Two (2) electrocoulometers shall be tested for the requirements of 3.4.1.2.2.
f. Fifty (50) electrocoulometers shall be tested for the requirements of 3.4.2.
g. Fifty (5) electrocoulometers shall be tested for the requirements of 3.4.3.
h. Thirty (30) electrocoulometers shall be tested for the requirements of 3.4.5.
i. Sixteen (16) electrocoulometers shall be tested for the requirements of 3.4.6.
j. Thirty (30) electrocoulometers shall be tested for the requirements of 3.4.7.

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k. Three hundred (300) electrocoulometers shall be tested for the requirements of 3.3.1.1 and 3.3.1.2 with:

(1) One hundred (100) devices tested at minus thirty-two degrees centigrade (-32°C).

(2) One hundred (100) devices tested at plus twenty-five degrees centigrade (+25°C).

(3) One hundred (100) devices tested at plus seventy-one degrees centigrade (+71°C).

4.2.3 Rejection. - If any electrocoulometer fails to comply with the applicable requirements, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.3.1 Lot formation. - The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sample procedure which is being applied to determine conformance with requirements. Lot numbering as required, shall be in accordance with MIL-STD-1168. Lot size shall be limited to a maximum of ten thousand (10,000) electrocoulometers or one (1) weeks production, which ever is less.

4.3.2 Examination. - The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accorance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using

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an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections shall be in accordance with 4.3.4.

4.3.2.1 Electrocoulometer (see dwg. 9276886).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Leads shorted	Visual	01001
102.	Length of body	Gage	01002
103.	Large diameter of body maximum (max)	Gage	01003
104.	Minimum lead length	Gage	01004
Minor:	AQL 1.0 percent		
201.	Undefined markings	Visual	01005
202.	Evidence of poor workmanship (see 3.9)	Visual	01006

4.3.3 Testing

4.3.3.1 Anode No. 1 Capacity, (see 3.3.1.1) Major Defect, Defect Code No. 10001. - Two hundred (200) electrocoulometers shall be randomly selected from each lot and tested for the specified requirements in accordance with 4.4.1. The lot shall be rejected if three (3) or more samples are classed defective.

4.3.3.2 Anode No. 2 Capacity, (See 3.3.1.2) Major Defect, Defect Code No. 10002. - Two hundred (200) electrocoulometers which have successfully passed the requirements of 3.3.1.1 shall be tested for the requirements of 3.3.1.2 in accordance with 4.4.2. The lot shall be rejected if one (1) or more samples are classed defective.

4.3.3.3 Cathode Lead Pull Test, (see 3.4.1.1.1) Major Defect, Code No. 10003. - Three (3) cathode lead - cup assemblies shall be tested for the specified requirements in accordance with 4.4.3. at the end of each hour of welding operations. The entire hour's production shall be rejected if one (1) or more assemblies are classed defective. Inspection requirements as stated shall comply to each welding machine and station used to produce cathode lead-cup assemblies.

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4.3.3.4 Cathode Lead Bend Test (see 3.4.1.1.2) Major defect, Code No. 10004. - Two (2) cathode lead - cup assemblies shall be tested for the specified requirements in accordance with 4.4.4, at the end of each hour of welding operations. The entire hour's production shall be rejected if one (1) or more assembleis are classed defective. Inspection requirements as stated shall apply to each welding machine and station used to produce cathode lead-cup assemblies.

4.3.3.5 Anode No. 2 Pull Test (see 3.4.1.2.1), Major Defect, Code No. 10005. - Three (3) electrocoulometers shall be tested for the specified requirements in accordance with 4.4.3 at the end of each hour's welding operations. The entire hour's production shall be rejected if one (1) or more assemblies are classed defective. Inspection requirements as stated shall apply to each welding machine and station used to weld anodes to the assembly.

4.3.3.6 Anode No. 2 Bend Test (see 3.4.1.2.2), Major Defect, Code No. 10006. - Two (2) electrocoulometers shall be tested for the specified requirements in accordance with 4.4.4 at the end of each hour's welding operations. The entire hour's production shall be rejected if one (1) or more assemblies are classed defective. Inspection requirements as stated shall apply to each welding machine and station used to weld anodes to the assembly.

4.3.3.7 Cathode Seal Examination (see 3.7.1), Major Defect, Code No. 10007. - Two hundred electrocoulometers shall be randomly selected from each lot and examined in accordance with 4.4.5 for the specified requirements. The lot shall be rejected if one (1) or more electrocoulometers are classed defective.

4.3.3.8 Anode Seal Examination (see 3.7.2), Major Defect, Code No. 10008. - Two hundred electrocoulometers shall be randomly selected from each lot and examined in accordance with 4.4.6 for the specified requirements. The lot shall be rejected if one (1) or more electrocoulometers are classed defective.

4.3.3.9 Hermetic Seal (see 3.4.2), Major Defect, Code No. 10009. - Twenty (20) electrocoulometers shall be randomly selected from each lot and tested in accordance with 4.4.7.2 for only the leakage requirements. The lot shall be rejected if one (1) or more electrocoulometers are declared defective.

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4.3.4 Inspection Equipment. - Inspection equipment required to perform the inspections prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test Methods And Procedures

4.4.1 Anode No. 1 Electrical Capacity. - Anode No. 1 of the electrocoulometer shall be tested for the specified capacity using approved test equipment capable of recording data during the plating period to show the specified outputs at the required times. The electrocoulometer shall be tested in the position requirements of 3.6. Each third shall be divided again into thirds and these thirds shall be tested at -32°C, + 25°C and + 71°C. Any sample that fails to comply with all of the specified requirements (3.3.1.1) shall be classed defective. Testing is destructive and test samples shall not be returned to the lot.

4.4.2 Anode No. 2 Electrical Capacity. - Anode No. 2 of the electrocoulometer shall be tested for the specified capacity using approved test equipment capable of recording data during the plating period to show the specified outputs at the required times. The electrocoulometer shall be tested in the position requirements of 3.6. Each third shall be divided again into thirds and these thirds shall be tested at -32°C, +25°C and +71°C. Positioning and temperature from the units tested in 4.3.3.1.1 and 4.4.1 for each unit shall be carried over into 4.3.3.1.2 and 4.4.2. Any sample that fails to comply with all of the specified requirements (3.3.1.2) shall be classed defective. Testing is destructive and test samples shall not be returned to the lot.

4.4.3 Pull Test. - The test specimen shall be secured in an approved fixture and pulled to destruction at the rate of three and five tenths plus or minus two tenths (3.5 ± 0.2) inches per minute. Units failing to comply with all of the specified requirements shall be classed defective. Testing is destructive and test samples shall not be returned to the lot.

4.4.4 Bend Test. - The test specimen shall be secured in an approved fixture and tested for the specified requirements. Units failing to comply with all of the specified requirements shall be classed defective. Testing is destructive and test samples shall not be returned to the lot.

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4.4.5 Cathode Seal Examination. - The electrocoulometer shall be examined for the specified visual defect requirements using an approved light source and ten power (10x) magnification device. If the electrocoulometer fails to comply with all of the specified requirements, it shall be classed defective. Testing is non-destructive and samples shall be returned to the lot.

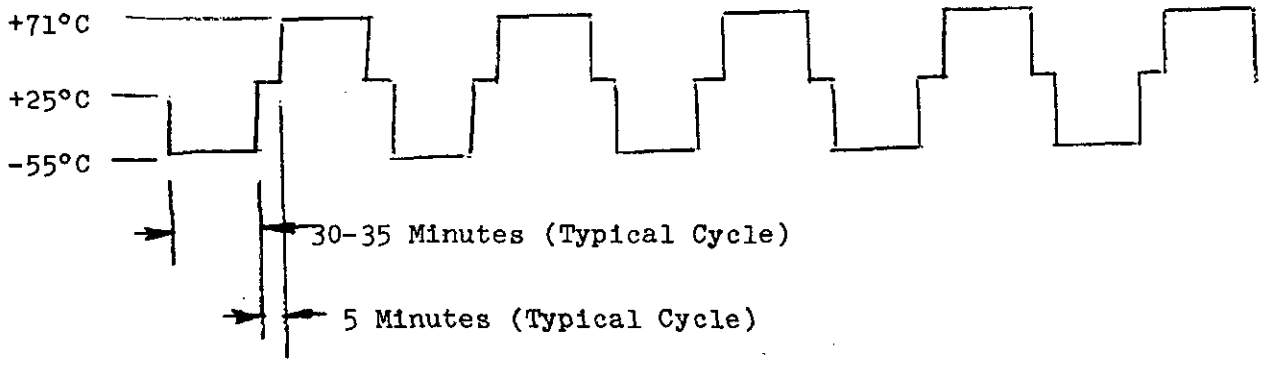
4.4.6 Anode Seal Examination. - The electrocoulometer shall be examined for the specified visual defect requirements by using the following procedure.

a. Clean the unit using approved techniques.
b. Dry the unit in an oven at seventy-one degrees centigrade (71°C) for fourteen plus five (14 + 5) hours.
c. Examine the unit using an approved light source and a ten power (10x) magnification device.

The unit shall then be classed defective if it fails to comply with all of the specified requirements after having been subjected to the above examination methods. Examination is non-destructive and test units shall be returned to the lot.

4.4.7 Hermetic Seal Tests

4.4.7.1 Hermetic Seal (First article procedure). - DUT shall be tested for the specified requirements in accordance with the attached graph of this paragraph. Free air must circulate around the DUT at all temperatures. The DUT, after the 5 thermal cycles are complete, shall be tested for leakage by drilling a hole near the cathode end of the case and draining the electrolyte from it; flooding the inside with helium and using a leak detector to test tive if it fails to comply with all of the specified requirements of 3.4.2. Testing is destructive and the DUT shall not be returned to the lot.



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4.4.7.2 Hermetic Seal (acceptance test procedure). - The electrocoulometer shall be tested by drilling a hole near the cathode end of the case and draining the electrolyte from it; flooding the inside with helium and using an approved leak detector to test for the presence of escaping gas. The unit shall be classed defective if it fails to comply with the leakage requirements. Testing is destructive and samples shall not be returned to the lot.

4.4.8 Extreme Temperature Storage. - The electrocoulometer shall be tested for its required 56 day test cycle in accordance with Test Method 112 of MIL-STD-331. Testing for capacity shall be performed in accordance with 4.4.1 and 4.4.2 and to the position requirements of 3.6. The device shall be classed defective if it fails to comply with all of the specified requirements of 3.4.3. Testing is destructive and the sample shall not be returned to the lot.

4.4.9 Mechanical Shock. - The electrocoulometer shall be tested for the specified shock requirements in accordance with Test Method 205C of MIL-STD-202. The positioning shall be in accordance with 3.6. Each third shall be shocked in one plane only. Each third shall be shocked in a plane different than the other groups of thirds. The electrocoulometer shall be tested for capacity after the shock test is completed in accordance with 4.4.1 and 4.4.2 with the position requirements of 3.6 applying. The device shall be classed defective if it fails to comply with all of the specified requirements (3.4.5). Testing is destructive and the sample shall not be returned to the lot.

4.4.10 Vibration. - The electrocoulometer shall be tested for the specified vibration requirements in accordance with Test Condition D, Test Method 204A of MIL-STD-202 (para. 4.5 excluded). Positioning shall be in accordance with 3.6. Each third shall be vibrated in one plane only. Each third shall be vibrated in a plane different from the other third groups. After vibration, capacities shall be tested in accordance with 4.4.1 and 4.4.2. Positioning requirements shall be in accordance with 3.6. The device shall be classed defective if it fails to comply with all of the specified requirements of 3.4.6. Testing is destructive and samples shall not be returned to the lot.

4.4.11 Altitude. - The electrocoulometer shall be tested for the altitude test requirements in an approved fixture. After altitude testing, capacities shall be tested in accordance with 4.4.1 and 4.4.2. Positioning requirements shall be in accordance with 3.6.

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The device shall be classed defective if it fails to comply with all of the specified requirements of 3.4.7. Testing is destructive and samples shall not be returned to the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C. - The electrocoulometer shall be preserved and packaged in accordance with MIL-STD-1169. The electrocoulometer shall be packaged into a container with section separators between each unit. Each anode shall be protected from shorting out in packing.

5.2 Packing

5.2.1 Level C. - The electrocoulometer, preserved and packaged as specified in 5.1.1 shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. - Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping. - When electrocoulometers of more than one lot are shipped together, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers. - The code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment deisgns submitted by the contractor to the Government for approval (see 6.3.3).

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6.3 Inspection equipment designs. - Inspection equipment designs are of two types: Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs. - SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.
b. A source control drawing or a specification control drawing as defined in MIL-STD-100.
c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor Designs. - Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is catalogued and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirement Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five-digit defect code number (or other specific

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identifying information) from Section 4 of this specification shall be referenced on each contractor design, together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval. - Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, Attn: SARPA-QA-T-T, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423)

6.4 Submission results of contractor conducted examinations and tests (see 3.18). - The contractor shall send his results to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-A-M
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 5999-A-056

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